5d. Vector Creation and Initialization

Martin Alfaro
PhD in Economics

INTRODUCTION

We continue presenting preliminary concepts for introducing the concept of mutation. The previous section distinguished between the use of `=` for assignments and mutations. Now, we'll deal with approaches to creating vectors.

Our presentation starts by outlining the process of initializing vectors, where memory is reserved without assigning initial values. We'll then discuss how to create vectors holding predefined values, such as zeros or ones. Finally, we show how to concatenate multiple vectors into new ones, through the `vcat` function.

INITIALIZING VECTORS

Creating an array involves two steps: reserving memory for holding its content and assigning initial values to its elements. When you don't intend to populate the array with values right away, it's more efficient to perform only the allocation step. This means reserving memory space, but without setting any initial values.

Technically, initializing entails creating an array filled with `undef` values. These values represent arbitrary content in memory at the moment of allocation. This is why, while `undef` displays concrete numbers when you output the array's content, they're meaningless and vary every time you initialize a new array.

There are two methods for creating vectors with `undef` values. The first one requires explicitly specifying the type and length of the array, which is accomplished via `Vector{<elements' type>}(undef, <length>)`. The second approach is based on the function `similar(x)`, which creates a vector with the same type and dimensions as an existing vector `x`.

'UNDEF' VALUES

```
x_length = 3

x             = Vector{Int64}(undef, x_length)   # `x` can hold `Int64` values, and is initialized with 3 undefined elements
```

```
julia> x
3-element Vector{Int64}:
 125590341097008
 125587347133984
 125587347134000
```

'SIMILAR'

```
y        = [3,4,5]

x          = similar(y)                          # `x` has the same type as `y`, which is Vector{Int64}(undef, 3)
```

```
julia> x
3-element Vector{Int64}:
 125590341768096
 125590286757888
 125590493769088
```

The example demonstrates that `undef` values don't follow any particular pattern. Moreover, these values vary in each execution, as they reflect any content held in RAM at the moment of allocation. In fact, a more descriptive way to call `undef` values would be **uninitialized values**.

CREATING VECTORS WITH GIVEN VALUES

In the following, we present several approaches to creating arrays holding predefined values.

VECTORS COMPRISING A RANGE

To generate a sequence of values, we can make use of **ranges**. Recall that the syntax for defining ranges is `<start>: <steps>: <stop>`, where `<steps>` specifies the increment between consecutive values.

In particular, we need the expression `collect(<range>)`. The function `collect` is necessary, since ranges describe the values to be generated, without materializing them. This behavior is consistent with a broader concept known as **lazy operations**, which will be presented later in this book. [1]

A VECTOR FILLED THROUGH A SEQUENCE

```
some_range = 2:5

x          = collect(some_range)
```

```
julia> x
4-element Vector{Int64}:
 2
 3
 4
 5
```

When a range is created, `<steps>` dictates the number of elements to be generated. Considering this, we can alternatively specify the number of elements to be stored through the syntax `<start> : 1/<number of elements> : <end>`. A more direct way is via the `range` function, whose syntax is `range(<start>, <end>, <number of elements>)`.

The following code snippet demonstrates the use of the `range` function, by generating five evenly spaced elements between 0 and 1.

RANGE

```
x = range(0, 1, 5)
```

```
julia> x
0.0:0.25:1.0
```

RANGE (KEYWORD ARGUMENTS)

```
x = range(start=0, stop=1, length=5)
```

```
julia> x
0.0:0.25:1.0
```

RANGE (KEYWORD ARGUMENTS)

```
x = range(start=0, length=5, stop=1)     # any order for keyword arguments
```

```
julia> x
0.0:0.25:1.0
```

VECTORS WITH SPECIFIC VALUES REPEATED

We can also build vectors of a given length, where every element holds the same value. Two common examples are `zeros` and `ones`, which respectively produce vectors filled with zeros and ones. By default, both functions create vectors whose elements have type `Float64`. You can override this behavior by providing the desired element type as the first argument.

ZEROS

```
length_vector = 3

x             = zeros(length_vector)
```

```
julia> x
3-element Vector{Float64}:
 0.0
 0.0
 0.0
```

ZEROS WITH TYPE

```
length_vector = 3

x             = zeros(Int, length_vector)
```

```
julia> x
3-element Vector{Int64}:
 0
 0
 0
```

ONES

```
length_vector = 3

x             = ones(length_vector)
```

```
julia> x
3-element Vector{Float64}:
 1.0
 1.0
 1.0
```

ONES WITH TYPE

```
length_vector = 3

x             = ones(Int, length_vector)
```

```
julia> x
3-element Vector{Int64}:
 1
 1
 1
```

To create vectors with Boolean values, Julia provides two convenient functions called `trues` and `falses`.

TRUES

```
length_vector = 3

x             = trues(length_vector)
```

```
julia> x
3-element BitVector:
 1
 1
 1
```

FALSES

```
length_vector = 3

x             = falses(length_vector)
```

```
julia> x
3-element BitVector:
 0
 0
 0
```

VECTORS WITH AN ARBITRARY VALUE REPEATED

We can define vectors whose elements all share the same specified value, without restricting this value to special cases like one or zero. This is achieved with the `fill` function.

FILL (SCALAR)

```
length_vector  = 3
filling_object = 1

x              = fill(filling_object, length_vector)
```

```
julia> x
3-element Vector{Int64}:
 1
 1
 1
```

FILL (VECTOR)

```
length_vector  = 3
filling_object = [1,2]

x              = fill(filling_object, length_vector)
```

```
julia> x
3-element Vector{Vector{Int64}}:
 [1, 2]
 [1, 2]
 [1, 2]
```

FILL (ONE-ELEMENT VECTOR)

```
length_vector  = 3
filling_object = [1]

x              = fill(filling_object, length_vector)
```

```
julia> x
3-element Vector{Vector{Int64}}:
 [1]
 [1]
 [1]
```

VECTORS FORMED WITH CONCATENATED VECTORS

There are several ways to construct a vector `z` that combines all elements of two vectors `x` and `y`. A straightforward way is by using `z = [x; y]`. While this works well for concatenating a small number of vectors, it becomes cumbersome when many vectors must be merged. Moreover, it's directly not viable when the number of vectors is unknown in advance.

In such cases, we can employ the `vcat` function. This concatenates all of its arguments into a single vector. When paired with the splat operator `...`, it can also operate on a single argument containing multiple vectors.

VCAT

```
x = [3,4,5]
y = [6,7,8]


z = vcat(x,y)
```

```
julia> x
6-element Vector{Int64}:
 3
 4
 ⋮
 7
 8
```

VCAT AND SPLATTING OPERATOR

```
x = [3,4,5]
y = [6,7,8]

A = [x, y]
z = vcat(A...)
```

```
julia> x
6-element Vector{Int64}:
 3
 4
 ⋮
 7
 8
```

VECTORS WITH REPEATED ELEMENTS FROM AN OBJECT

Closely related to `fill` is the `repeat` function. This takes a *collection* as an input, concatenating their elements repeatedly a specified number of times. The `repeat` function **necessarily requires an array as its input**, throwing an error if a scalar is passed.

REPEAT (VECTOR)

```
nr_repetitions     = 3
elements_to_repeat = [1,2]

x                  = repeat(elements_to_repeat, nr_repetitions)
```

```
julia> x
6-element Vector{Int64}:
 1
 2
 ⋮
 1
 2
```

REPEAT (VECTOR - SINGLE ELEMENT)

```
nr_repetitions     = 3
elements_to_repeat = [1]

x                  = repeat(elements_to_repeat, nr_repetitions)
```

```
julia> x
3-element Vector{Int64}:
 1
 1
 1
```

REPEAT (SCALAR - NOT ALLOWED)

```
nr_repetitions   = 3
vector_to_repeat = 1

x                = repeat(vector_to_repeat, nr_repetitions)
```

```
ERROR: MethodError: no method matching repeat(::Int64, ::Int64)
```

Note that the behavior of `repeat` differs from `fill` function, since the latter repeats the same object without concatenating its elements. In fact, the output of `repeat` is equivalent to creating a vector with `fill` and then using `vcat` to concatenate its elements.

'REPEAT' WITH VECTOR INPUT

```
nr_repetitions     = 3
elements_to_repeat = [1,2]

x                  = repeat(elements_to_repeat, nr_repetitions)
```

```
julia> x
6-element Vector{Int64}:
 1
 2
 ⋮
 1
 2
```

'FILL' WITH VECTOR INPUT

```
length_vector      = 3
filling_object     = [1,2]

x                  = fill(filling_object, length_vector)
```

```
julia> x
3-element Vector{Vector{Int64}}:
 [1, 2]
 [1, 2]
 [1, 2]
```

'FILL' + 'VCAT'

```
length_vector      = 3
filling_object     = [1,2]

temp               = fill(filling_object, length_vector)
x                  = vcat(temp...)
```

```
julia> x
6-element Vector{Int64}:
 1
 2
 ⋮
 1
 2
```

ADDING, REMOVING, AND REPLACING ELEMENTS (*OPTIONAL*)

> ⚠️ **Warning!**
>
> This subsection requires knowledge of a few **concepts we haven't discussed yet**. As such, it's marked as optional.
>
> One such concept is that of **in-place functions**, which will be introduced later. These functions are identified by the symbol `!` appended to the function's name. The symbol is simply a convention chosen by developers, not a syntax requirement. It indicates that the function modifies at least one of its arguments.
>
> Another concept introduced is that of **pairs**, which will be thoroughly examined in a future section too. For the purposes of this subsection, it's sufficient to know that pairs are written in the form `a => b`. In our applications, `a` will represent a given value and `b` denote its corresponding replacement value.

Next, we demonstrate how to add, remove, and replace elements within a vector. We begin by presenting methods for adding a single element.

ADD SCALAR AT END

```
x                = [3,4,5]
element_to_insert = 0


push!(x, element_to_insert)                # add 0 as last element - faster
```

```
julia> x
4-element Vector{Int64}:
 3
 4
 5
 0
```

ADD ONE ELEMENT AT START

```
x                = [3,4,5]
element_to_insert = 0


pushfirst!(x, element_to_insert)           # add 0 as first element - slower
```

```
julia> x
4-element Vector{Int64}:
 0
 3
 4
 5
```

ADD ONE ELEMENT AT SPECIFIC INDEX

```
x                = [3,4,5]
element_to_insert = 0
at_index         = 2

insert!(x, at_index, element_to_insert)     # add 0 at index 2
```

```
julia> x
4-element Vector{Int64}:
 3
 0
 4
 5
```

ADD MULTIPLE ELEMENTS AT END

```
x                 = [3,4,5]
vector_to_insert  = [6,7]


append!(x, vector_to_insert)                # add 6 and 7 as last elements
```

```
julia> x
5-element Vector{Int64}:
 3
 4
 5
 6
 7
```

The function `push!` is particularly helpful to collect outputs in a vector. Since it doesn't require prior knowledge of how many elements will be stored, the vector can grow dynamically as new results are added. Notice that adding elements at the end with `push!` is generally faster than inserting them at the beginning with `pushfirst!`.

Analogous functions exist to remove elements, as shown below.

REMOVE LAST ELEMENT

```
x                 = [5,6,7]


pop!(x)                             # delete last element
```

```
julia> x
2-element Vector{Int64}:
 5
 6
```

REMOVE FIRST ELEMENT

```
x                 = [5,6,7]


popfirst!(x)                        # delete first element
```

```
julia> x
2-element Vector{Int64}:
 6
 7
```

REMOVE ONE ELEMENT AT INDEX

```
x                 = [5,6,7]
index_of_removal  = 2

deleteat!(x, index_of_removal)     # delete element at index 2
```

```
julia> x
2-element Vector{Int64}:
 5
 7
```

REMOVE MULTIPLE ELEMENTS AT INDEX

```
x                 = [5,6,7]
indices_of_removal = [1,3]

deleteat!(x, indices_of_removal)   # delete elements at indices 1 and 3
```

```
julia> x
1-element Vector{Int64}:
 6
```

By analogy with the behavior of `deleteat!`, it's also possible to specify which elements should be retained.

KEEP SINGLE ELEMENT

```
x              = [5,6,7]
index_to_keep  = 2

keepat!(x, index_to_keep)
```

```
julia> x
1-element Vector{Int64}:
 6
```

KEEP MULTIPLE ELEMENTS

```
x              = [5,6,7]
indices_to_keep = [2,3]

keepat!(x, index_to_keep)
```

```
julia> x
1-element Vector{Int64}:
 6
```

Finally, specific values can be replaced with new ones. This can be done by either creating a new copy using `replace` or by updating the original vector with `replace!`.

Both functions make use of pairs `a => b`, where `a` denotes the value to be replaced and `b` specifies its replacement. Note that these functions perform substitutions based on values, not on indices.

REPLACE SINGLE VALUE (UPDATE)

```
x = [3,3,5]

replace!(x, 3 => 0)              # in-place (it updates x)
```

```
julia> x
3-element Vector{Int64}:
 0
 0
 5
```

REPLACE MULTIPLE VALUES (UPDATE)

```
x = [3,3,5]

replace!(x, 3 => 0, 5 => 1)      # in-place (it updates x)
```

```
julia> x
3-element Vector{Int64}:
 0
 0
 1
```

REPLACE SINGLE VALUE (COPY)

```
x = [3,3,5]

y = replace(x, 3 => 0)           # new copy
```

```
julia> y
3-element Vector{Int64}:
 0
 0
 5
```

REPLACE MULTIPLE VALUES (COPY)

```
x = [3,3,5]

y = replace(x, 3 => 0, 5 => 1)   # new copy
```

```
julia> y
3-element Vector{Int64}:
 0
 0
 1
```

FOOTNOTES

[1.] Lazy operations specify how a computation should proceed, but postpone producing concrete results until they're explicitly required (either because you request them or because another computation depends on them.) They become especially valuable when combined with other operations, since it can fuse operations and therefore eliminate the need to store intermediate results altogether.